multi_agent_financial_analysis

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1 Multi-Agent Financial Analysis System

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This notebook is the canonical, executable report for an agentic investment-research system. It accepts a stock ticker and user research question, uses LangGraph to route a bounded run to market, financial, news, or comprehensive specialists, evaluates and revises the complete report, and stores reusable lessons for later runs.

Repository: AAI-520 Final Project

> Research support only. The output is not personalized investment advice and does not guarantee future performance.

1.1 1. Project objectives and implemented capabilities

The project demonstrates four autonomous agent functions:

1. **Plan** research objectives, questions, and tool selection using prior-run memory as context.
2. **Use tools** to retrieve Yahoo Finance market/fundamental data, SEC EDGAR facts and filings, and multi-source company news.
3. **Reflect and evaluate** data quality, report grounding, and specialist-analysis completeness.
4. **Learn across runs** by persisting bounded, timestamped research lessons.

It also implements all three required workflow patterns:

- **Prompt chaining:** ingest → preprocess → classify → extract → summarize.
- **Routing:** the user request selects market-only, financial-only, news-only, or comprehensive research; news articles also route to earnings, market, and general-news specialists.
- **Evaluator–optimizer:** article analysis uses a four-part rubric, while the complete report is scored for completeness, grounding, reasoning, consistency, sources, and clarity and revised through a bounded LangGraph loop.

The refactored modules preserve the original notebook's planning, tool execution, validation, reflection, reporting, memory, SEC EDGAR retrieval, 10-year price graph, multi-source news analysis, specialist routing, and iterative optimization. The updated Google News RSS fallback and retry-aware HTTP behavior are included as well.

1.2 2. Implemented system architecture

The research workflow keeps provider adapters, provider-independent tools, agents, workflows, memory, and presentation concerns separate. This makes each stage inspectable and independently

testable.

```
flowchart LR
    U[User ticker and research question] --> P[Planner]
    L[Research memory] --> P
    P --> RQ{Request route}
    RQ -->|Market| MA[Market specialist]
    RQ -->|Financial| FA[Financial specialist]
    RQ -->|News| C[News prompt chain]
    RQ -->|Comprehensive| MA
    RQ -->|Comprehensive| FA
    RQ -->|Comprehensive| C
    Y[Yahoo Finance] --> MA
    Y --> FA
    S[SEC EDGAR] --> FA
    Y --> C
    N[NewsAPI] --> C
    G[Google News RSS fallback] --> C
    C --> R[Specialist router]
    R --> E[Earnings specialist]
    R --> M[Market specialist]
    R --> W[News specialist]
    E --> V[Evaluator / optimizer]
    M --> V
    W --> V
    MA --> Z[Synthesizer]
    FA --> Z
    V --> Z
    Z --> Q{Six-part report evaluator}
    Q -->|Below threshold| O[Report optimizer]
    O --> Q
    Q -->|Pass or revision limit| F[Final report and visuals]
    F --> L
```

1.3 3. Environment setup

Create and activate a virtual environment, install `requirements.txt`, start Ollama, and copy `.env.example` to `.env`. At minimum:

- `OLLAMA_MODEL=llama3.2` and a running `ollama serve` process;
- `SEC_USER_AGENT` containing an application name and monitored contact email;
- optional `NEWS_API_KEY` (Yahoo remains primary and Google RSS is a fallback);
- optional `DEFAULT_TICKER`, `RESEARCH_REQUEST`, `MAX_NEWS_ARTICLES`, `EVALUATION_SCORE_THRESHOLD`, `MAX_OPTIMIZER_ITERATIONS`, `REPORT_SCORE_THRESHOLD`, and `MAX_REPORT_REVISIONS`.

Credentials are read only from environment variables and are never displayed below.

```
Local environment detected; start Ollama separately with ollama serve.
```

```
Project root: /Users/brandonwirgau/Projects/AAI/520/Final Project
Ticker: AAPL
Research request: How do market performance, financial condition, SEC evidence,
and recent news affect the company's current risk picture?
NewsAPI configured: False
SEC identity configured: True
```

1.4 4. Agent design, data sources, and tool use

Component	Implemented responsibility
Planner Agent	Uses the ticker, registered tools, and the last three relevant memories to produce objectives, questions, and an allow-listed tool plan.
Tool Executor	Runs only registered tools and isolates provider errors so one failure does not terminate the research run.
Market Specialist / Tools	Retrieve price and volume history; compute one-month, three-month, and one-year returns, volatility, drawdown, 52-week range, and average volume.
Financial Specialist / Tools	Retrieve income, cash-flow, and balance-sheet statements; compute revenue/earnings growth and analyze cash, debt, net debt, assets, liabilities, and equity.
SEC EDGAR Tool	Resolves CIK, reads official company facts, and lists recent 10-K, 10-Q, and 8-K filings with source metadata.
News Prompt Chain	Combines Yahoo and NewsAPI; invokes Google RSS when primary coverage is missing or sparse; deduplicates and structures articles.
Specialist Router	Sends classified items to earnings, market, or general-news specialists.
Evaluator / Optimizer	Scores observations, interpretation, uncertainty, and evidence breadth; revises until the threshold or iteration limit.
Synthesis Agent	Produces a grounded Markdown report spanning market, financial, filing, and news evidence.
Memory Curator / Store	Saves bounded, timestamped lessons using atomic file replacement; memory is planning context, never current evidence.

FRED and Alpha Vantage packages remain deliberate extension points. The report does not claim they are integrated data sources.

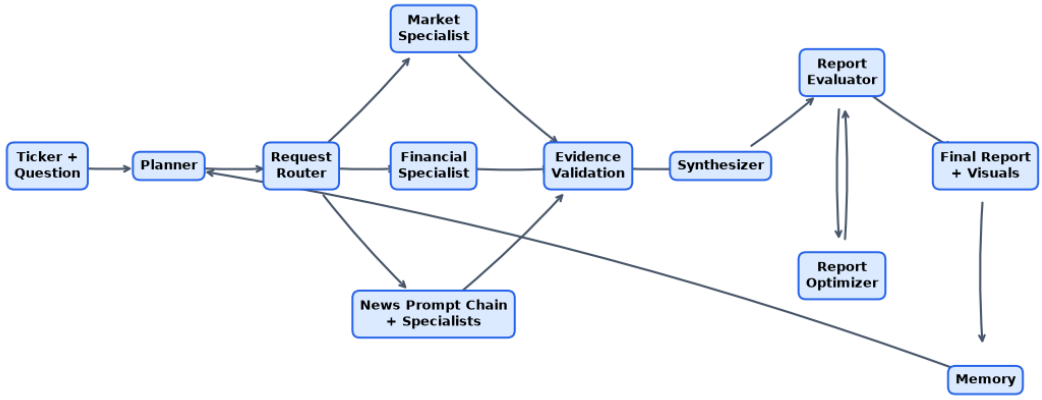
```
              tool                                          role
0       price_data   1m/3m/1y performance, risk, range, and volume
```

```
1      company_info       Profile, valuation, margins, and dividend
2        financials                          Income-statement trends
3         cash_flow  Operating cash flow, free cash flow, and capex
4     balance_sheet     Cash, debt, assets, liabilities, and equity
5  financial_growth             Revenue, net-income, and EPS growth
6         sec_edgar               Official facts and recent filings
```

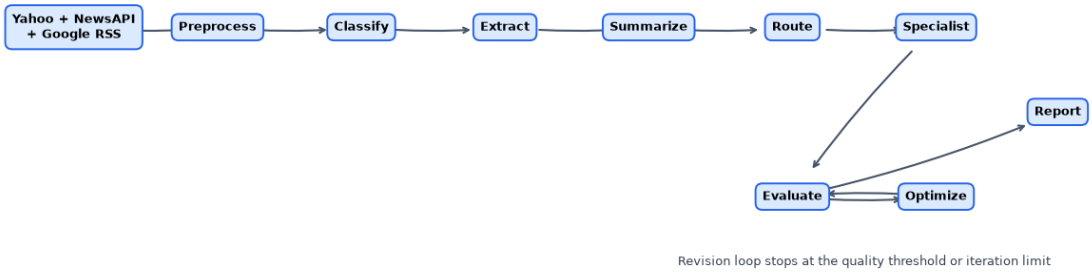
1.5 5. Workflow diagrams

These diagrams are generated by the project's reporting module, so the submitted report reflects the implemented code rather than a disconnected illustration.

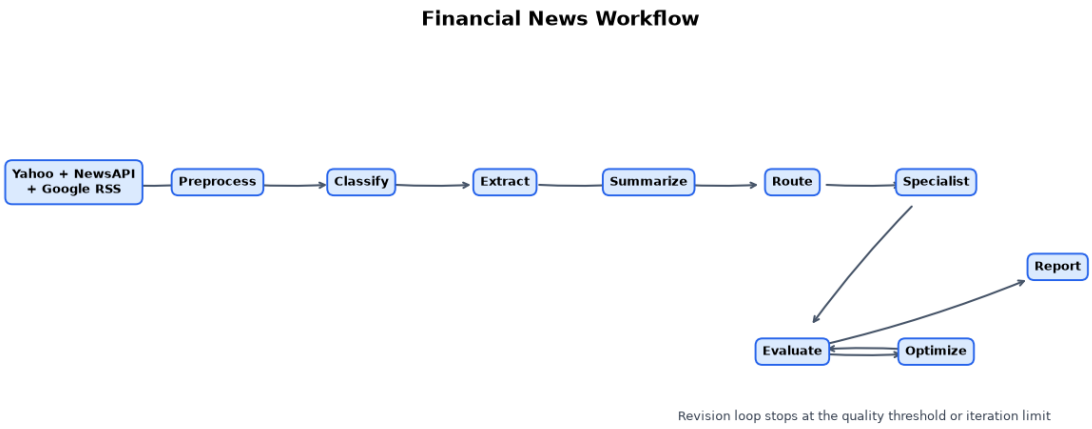
Multi-Agent Financial Analysis — System Architecture



Financial News Workflow



Revision loop stops at the quality threshold or iteration limit



1.6 6. Prompt chaining, routing, and evaluator–optimizer logic

The updated news workflow uses provider isolation and explicit fallback logic:

```
flowchart LR
    Y[Yahoo Finance] --> D[Deduplicate and newest-sort]
    N[NewsAPI] --> D
    Y --> F{Coverage sparse?}
    N --> F
    F -->|Yes| G[Google News RSS]
    G --> D
    D --> P[Preprocess]
    P --> C[Classify]
    C --> X[Extract facts, figures, events]
    X --> S[Structured summary]
    S --> R{Route by category}
    R --> E[Earnings]
    R --> M[Market]
    R --> W[General news]
    E --> V[4 x 25-point evaluation]
    M --> V
    W --> V
    V -->|Below threshold| O[Optimizer]
    O --> V
    V -->|Pass or max iterations| Z[Synthesis]
```

Provider failures are retained in `source_errors`; original and optimized analyses plus every evaluation score remain in workflow state for auditability. NewsAPI and RSS use retry-enabled HTTP sessions for transient 429/5xx and connection failures.

1.7 7. End-to-end investment research run

The next cell performs two live LangGraph runs for the same request. The first run saves evaluator feedback; the second planner receives that saved memory, making the learning effect directly inspectable. Each run:

1. accepts and classifies the research request;
2. conditionally routes to market, financial, news, or all specialists;
3. retrieves Yahoo Finance, SEC, and prompt-processed news evidence as selected;
4. validate observations;
5. reflect on evidence quality;
6. synthesize the report;
7. scores the complete report across six criteria;
8. revises and re-evaluates below-threshold reports up to a configured limit; and
9. saves reusable evaluator feedback when memory is enabled.

A running local Ollama model and network access are required for a complete live result. Missing optional providers are recorded rather than treated as fatal.

```
FIRST RUN - establish evidence and save feedback
Planning and routing the research request

Running all selected research specialists
Running market tool: company_info

Running market tool: price_data
Running financial tool: balance_sheet

Running financial tool: cash_flow
Running financial tool: company_info

Running financial tool: financial_growth

Running financial tool: financials
Running financial tool: sec_edgar

Processing news article 1/8

Processing news article 2/8

Processing news article 3/8

Processing news article 4/8

Processing news article 5/8

Processing news article 6/8

Processing news article 7/8

Processing news article 8/8

Validating specialist observations
Reflecting on evidence quality

Synthesizing the draft report
```

```
Scoring the complete report
Revising the complete report (iteration 1)

Scoring the complete report
Finalizing the evaluated report
Saving evaluator feedback for a future run


SECOND RUN - plan with the saved feedback
Planning and routing the research request

Running all selected research specialists
Running market tool: company_info

Running market tool: price_data
Running financial tool: balance_sheet
Running financial tool: cash_flow
Running financial tool: company_info
Running financial tool: financial_growth
Running financial tool: financials
Running financial tool: sec_edgar

Processing news article 1/8

Processing news article 2/8

Processing news article 3/8

Processing news article 4/8

Processing news article 5/8

Processing news article 6/8

Processing news article 7/8

Processing news article 8/8

Validating specialist observations
Reflecting on evidence quality

Synthesizing the draft report

Scoring the complete report
Revising the complete report (iteration 1)

Scoring the complete report
Finalizing the evaluated report
```

1.7.1 7.1 Structured planning and quality artifacts


```
{
  "symbol": "AAPL",
  "research_request": "How do market performance, financial condition, SEC
evidence, and recent news affect the company's current risk picture?",
  "request_route": "comprehensive",
```

```
  "selected_specialists": [
    "market",
    "financial",
    "news"
  ],
  "plan": {
    "objectives": [
      "Understand how market performance affects the company's current risk
picture",
      "Analyze the impact of financial condition on the company's current risk
picture",
      "Examine the effect of SEC evidence on the company's current risk
picture",
      "Assess the influence of recent news on the company's current risk
picture"
    ],
    "tools": [
      "balance_sheet",
      "cash_flow",
      "company_info",
      "financial_growth",
      "financials",
      "price_data",
      "sec_edgar"
    ],
    "questions": [
      "What is the current market performance of AAPL and how does it impact the
company's risk picture?",
      "How does the company's financial condition (e.g., cash flow, debt)
reflect its current risk profile?",
      "What SEC evidence (e.g., EDGAR filings) can be used to inform the
company's risk assessment?",
      "What recent news stories or events may be relevant to the company's
current risk picture?",
      "Which material claims can be linked to filing-level citations or provider
source URLs?"
    ],
    "research_request": "How do market performance, financial condition, SEC
evidence, and recent news affect the company's current risk picture?",
    "route": "comprehensive",
    "specialists": [
      "market",
      "financial",
      "news"
    ],
    "memory_context": [
      {
        "date": "2026-10-17T05:02:14.249666+00:00",
```

```
        "symbol": "AAPL",
        "summary": "Here are the reusable research lessons from this AAPL
run:\n\n1. **Evidence coverage**: Ensure that primary-source evidence (e.g., SEC
EDGAR) is used to verify financial metrics, but be cautious when comparing data
from different providers and reporting periods.\n2. **Data-quality problems**:
Verify data quality by cross-checking values from different providers and
reporting periods to avoid misinterpretation.\n3. **Improvements for future
runs**:\n* Include at least two evidence-based observations to strengthen the
analysis.\n* Verify the linked sources for news articles to ensure accuracy.\n*
Use provider-supplied data for market performance and news, but be cautious of
potential biases and limitations.\n* Consider using multiple sources to verify
financial metrics and balance-sheet data.\n* Improve financial reasoning and
event-significance analysis to strengthen the analysis.\n* Consider using more
advanced analytics and modeling techniques to better understand the company's
risk picture.",
        "weaknesses": [
          "Substantive financial reasoning or event-significance analysis
remains limited"
        ]
      }
    ]
  },
  "specialist_analyses": {
    "market": {
      "performance": {
        "1m_return_percent": 1.92,
        "3m_return_percent": 5.26,
        "1y_return_percent": 36.64
      },
      "risk": {
        "annualized_volatility_percent": 24.4,
        "maximum_drawdown_percent": -13.8
      },
      "range_and_volume": {
        "52_week_high": 348.15,
        "52_week_low": 242.76,
        "latest_volume": 59895300.0,
        "average_volume": 48103449.4
      }
    },
    "financial": {
      "growth": {
        "revenue_growth_percent": 6.43,
        "net_income_growth_percent": 19.5,
        "diluted_eps_growth_percent": 22.7,
        "latest_period": "2025-09-30 00:00:00",
        "prior_period": "2024-09-30 00:00:00"
      },
```

```
      "cash_and_debt": {
        "cash": 54697000000.0,
        "debt": 98657000000.0,
        "net_debt": 43960000000.0,
        "cash_to_debt_ratio": 0.5544,
        "total_assets": 359241000000.0,
        "total_liabilities": 285508000000.0,
        "stockholders_equity": 73733000000.0
      },
      "balance_sheet_trends": {
        "cash_growth_percent": -16.07,
        "debt_growth_percent": -7.48,
        "assets_growth_percent": -1.57,
        "liabilities_growth_percent": -7.31,
        "equity_growth_percent": 29.47,
        "latest_period": "2025-09-30 00:00:00",
        "prior_period": "2024-09-30 00:00:00"
      },
      "has_sec_evidence": true
    },
    "news": {
      "article_count": 8,
      "average_score": 100.0,
      "average_structural_score": 100.0,
      "average_substantive_reasoning_score": 50.0,
      "routing_counts": {
        "market": 6,
        "news": 2
      }
    }
  },
  "validation": [],
  "reflection": {
    "strengths": [
      "Evidence coverage",
      "Data-quality problems",
      "Improvements for future runs"
    ],
    "weaknesses": [
      "Substantive financial reasoning or event-significance analysis remains
limited"
    ],
    "missing_information": [],
    "suspicious_values": [],
    "follow_up_questions": [
      "Include at least two evidence-based observations",
      "Verify the linked sources for news articles",
      "Use provider-supplied data for market performance and news",
```

```
      "Consider using multiple sources to verify financial metrics and balance-
sheet data"
    ],
    "deterministic_issues": []
  },
  "report_evaluation": {
    "score": 90.0,
    "passed": true,
    "threshold": 75.0,
    "minimum_criterion_score": 60.0,
    "criteria": {
      "completeness": 100.0,
      "grounding": 100.0,
      "reasoning": 60.0,
      "consistency": 100.0,
      "sources": 80.0,
      "clarity": 100.0
    },
    "feedback": [
      "Improve reasoning; current score is 60/100."
    ],
    "validation_issues": []
  },
  "report_revision_history": [
    {
      "iteration": 1,
      "before_score": 82.5,
      "after_score": 90.0,
      "feedback": [
        "Improve reasoning; current score is 15/100.",
        "No report can pass while these criteria are below 60/100: reasoning."
      ]
    }
  ],
  "graph_trace": [
    "plan_request",
    "comprehensive_research",
    "validate_observations",
    "reflect",
    "synthesize",
    "evaluate_report",
    "revise_report",
    "evaluate_report",
    "finalize"
  ]
}
```


```
                                    research_request selected_route  \
```

```
0  How do market performance, financial condition…  comprehensive

      selected_specialists
0  market, financial, news
```

1.7.2 7.2 Collected market, fundamental, and SEC evidence

The main report shows compact evidence tables so the important values and reporting periods remain reviewable. The complete observation object is retained in the companion audit HTML generated in the appendix. Provider errors appear alongside successful results instead of being hidden.

Market and company snapshot

	source	metric	value
0	price_data	latest_price	343.46
1	price_data	1m_return_percent	1.92
2	price_data	3m_return_percent	5.26
3	price_data	1y_return_percent	36.64
4	price_data	annualized_volatility_percent	24.4
5	price_data	maximum_drawdown_percent	-13.8
6	price_data	52_week_high	348.15
7	price_data	52_week_low	242.76
8	price_data	latest_volume	59895300.0
9	price_data	average_volume	48103449.4
10	company_info	longName	Apple Inc.
11	company_info	sector	Technology
12	company_info	industry	Consumer Electronics
13	company_info	marketCap	5012516765696
14	company_info	enterpriseValue	5034461888512
15	company_info	trailingPE	39.20776
16	company_info	forwardPE	35.85982
17	company_info	dividendYieldPercent	0.32

Latest income-statement and cash-flow observations

	source	metric	period	value
0	financials	Total Revenue	2025-09-30 00:00:00	4.161610e+11
1	financials	Operating Income	2025-09-30 00:00:00	1.330500e+11
2	financials	Net Income	2025-09-30 00:00:00	1.120100e+11
3	financials	EBITDA	2025-09-30 00:00:00	1.447480e+11
4	financials	Diluted EPS	2025-09-30 00:00:00	7.460000e+00
5	cash_flow	Operating Cash Flow	2025-09-30 00:00:00	1.114820e+11
6	cash_flow	Free Cash Flow	2025-09-30 00:00:00	9.876700e+10
7	cash_flow	Capital Expenditure	2025-09-30 00:00:00	-1.271500e+10

Balance sheet and deterministic growth

```
                     metric                                    period  \
0                      cash                       2025-09-30 00:00:00
1                      debt                       2025-09-30 00:00:00
2                  net_debt                       2025-09-30 00:00:00
3          cash_to_debt_ratio                     2025-09-30 00:00:00
4               total_assets                      2025-09-30 00:00:00
5           total_liabilities                     2025-09-30 00:00:00
6         stockholders_equity                     2025-09-30 00:00:00
7      revenue_growth_percent  2024-09-30 00:00:00 to 2025-09-30 00:00:00
8   net_income_growth_percent  2024-09-30 00:00:00 to 2025-09-30 00:00:00
9  diluted_eps_growth_percent  2024-09-30 00:00:00 to 2025-09-30 00:00:00

          value
0  5.469700e+10
1  9.865700e+10
2  4.396000e+10
3  5.544000e-01
4  3.592410e+11
5  2.855080e+11
6  7.373300e+10
7  6.430000e+00
8  1.950000e+01
9  2.270000e+01
```

SEC company facts and filing periods

```
                 fact         value        unit  fiscal_year fiscal_period  \
0             Revenue  3.643570e+11         USD         2026            Q3
1          Net Income  1.014640e+11         USD         2026            Q3
2    Operating Income  1.224320e+11         USD         2026            Q3
3  Operating Cash Flow  1.169960e+11         USD         2026            Q3
4              Assets  3.832660e+11         USD         2026            Q3
5         Liabilities  2.757460e+11         USD         2026            Q3
6  Stockholders Equity  1.075200e+11         USD         2026            Q3
7         Diluted EPS  6.880000e+00  USD/shares         2026            Q3

   form  period_end       filed             accession
0  10-Q  2026-06-27  2026-07-31  0000320193-26-000020
1  10-Q  2026-06-27  2026-07-31  0000320193-26-000020
2  10-Q  2026-06-27  2026-07-31  0000320193-26-000020
3  10-Q  2026-06-27  2026-07-31  0000320193-26-000020
4  10-Q  2026-06-27  2026-07-31  0000320193-26-000020
5  10-Q  2026-06-27  2026-07-31  0000320193-26-000020
6  10-Q  2026-06-27  2026-07-31  0000320193-26-000020
7  10-Q  2026-06-27  2026-07-31  0000320193-26-000020

   form filing_date report_date             accession  \
0  10-Q  2026-07-31  2026-06-27  0000320193-26-000020
```

```
1   8-K  2026-07-30  2026-07-30  0000320193-26-000018
2  10-Q  2026-05-01  2026-03-28  0000320193-26-000013
3   8-K  2026-04-30  2026-04-30  0000320193-26-000011
4   8-K  2026-04-20  2026-04-17  0001140361-26-015711
5   8-K  2026-02-24  2026-02-24  0001140361-26-006577
6  10-Q  2026-01-30  2025-12-27  0000320193-26-000006
7   8-K  2026-01-29  2026-01-29  0000320193-26-000005

                                          filing_url
0  https://www.sec.gov/Archives/edgar/data/320193…
1  https://www.sec.gov/Archives/edgar/data/320193…
2  https://www.sec.gov/Archives/edgar/data/320193…
3  https://www.sec.gov/Archives/edgar/data/320193…
4  https://www.sec.gov/Archives/edgar/data/320193…
5  https://www.sec.gov/Archives/edgar/data/320193…
6  https://www.sec.gov/Archives/edgar/data/320193…
7  https://www.sec.gov/Archives/edgar/data/320193…
```

Complete raw observations are preserved in the companion audit HTML.

1.7.3 7.3 Detailed news analysis

This table surfaces source provenance, routing decisions, final evaluator scores, and optimizer activity. The structural score measures field completeness and source traceability; the separate substantive score measures financial reasoning and event significance. A 100 structural score must not be interpreted as perfect financial analysis. Complete per-article records remain in the companion audit HTML.

```
   article           source               published_at  \
0        1  Google News RSS  2026-10-16T13:14:35+00:00
1        2  Google News RSS  2026-10-16T11:37:30+00:00
2        3  Google News RSS  2026-10-16T11:18:12+00:00
3        4  Google News RSS  2026-10-16T08:39:31+00:00
4        5  Google News RSS  2026-10-16T08:39:31+00:00
5        6  Google News RSS  2026-10-16T08:39:31+00:00
6        7  Google News RSS  2026-10-15T13:38:15+00:00
7        8  Google News RSS  2026-10-13T12:00:02+00:00

                                          headline category  \
0  Apple Stock Hits Record High as iPhone Duo Pre…   market
1  Apple Inc. Price, AAPL Price, Live Charts, and…   market
2  Dan Ives Says Apple-Meta AI Partnership Could …     news
3  Apple Inc. $AAPL Stock Sold by Investors Resea…     news
4  Apple Inc. $AAPL Stock Sold by Persium Advisor…   market
5  Apple Inc. $AAPL Stock: Wealth Enhancement Tru…   market
6  Apple (AAPL): Can Longer iPhone Ownership Stil…   market
7  Apple Inc. (AAPL) is Attracting Investor Atten…   market

   classification_confidence       company  \
```

```
0                       0.8         Apple
1                       0.9    Apple Inc.
2                       0.9          None
3                       0.9    Apple Inc.
4                       0.9    Apple Inc.
5                       0.8    Apple Inc.
6                       0.8  Apple (AAPL)
7                       0.8    Apple Inc.

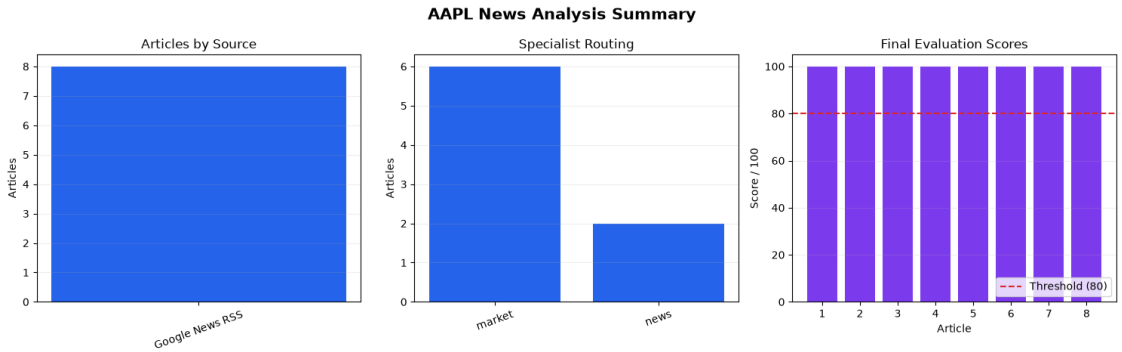
                                          event   event_dates sentiment  \
0                    iPhone Duo pre-orders sell out            []     mixed
1                                             Price            []     mixed
2                                              None            []   neutral
3                   Sold by Investors Research Corp            []  negative
4                                              Sold  [2023-11-15]  negative
5  Wealth Enhancement Trust Services Inc. Increas…            []     mixed
6  Can Longer iPhone Ownership Still Produce More…  [2023-12-01]     mixed
7                     Attracting Investor Attention            []     mixed

  specialist  structural_score  structural_passed  \
0    market               100               True
1    market               100               True
2      news               100               True
3      news               100               True
4    market               100               True
5    market               100               True
6    market               100               True
7    market               100               True

   substantive_reasoning_score  substantive_reasoning_passed  \
0                           50                         False
1                           50                         False
2                           50                         False
3                           50                         False
4                           50                         False
5                           50                         False
6                           50                         False
7                           50                         False

   optimizer_revisions                                                 url
0                    0  https://news.google.com/rss/articles/CBMizwFBV…
1                    1  https://news.google.com/rss/articles/CBMikgFBV…
2                    0  https://news.google.com/rss/articles/CBMirwJBV…
3                    0  https://news.google.com/rss/articles/CBMiswFBV…
4                    0  https://news.google.com/rss/articles/CBMirwFBV…
5                    1  https://news.google.com/rss/articles/CBMi0AFBV…
6                    0  https://news.google.com/rss/articles/CBMioAFBV…
7                    0  https://news.google.com/rss/articles/CBMioAFBV…
```

```
{
  "pipeline_stages": [
    "ingest",
    "preprocess",
    "classify",
    "extract",
    "summarize",
    "route",
    "specialist",
    "evaluate",
    "optimize"
  ],
  "retrieved_at": "2026-10-17T05:02:47.807462+00:00",
  "retrieved_counts": {
    "Yahoo Finance": 0,
    "NewsAPI": 0,
    "Google News RSS": 8
  },
  "source_errors": {},
  "source_counts": {
    "Google News RSS": 8
  },
  "routing_counts": {
    "market": 6,
    "news": 2
  },
  "average_structural_score": 100.0,
  "average_substantive_reasoning_score": 50.0,
  "score_scope": "Structural completeness and source traceability are scored
separately from substantive financial reasoning and event significance.",
  "passed_count": 8
}
```

1.8 8. Final research report

The model-generated report uses only collected observations and explicitly distinguishes evidence, interpretation, risk, uncertainty, and missing data.

1.9 Company Overview

- Company Name: Apple Inc.
- Symbol: AAPL
- Sector: Technology
- Industry: Consumer Electronics
- Country: United States

1.10 Price Performance

1.11 Valuation and Profitability

1.12 Financial Performance

1.13 Cash Flow

- Cash/Debt Ratio: 0.5544
- Assets: $359,241,000,000

1.14 Recent News and Catalysts

- News Source: Google News RSS
- Publisher: Benzinga, Coinbase, MarketBeat, Yahoo Finance
- Publication Date:
  - Apple Inc. $AAPL Stock Sold by Investors Research Corp - MarketBeat: 2026-10-16T08:39:31+00:00
  - Dan Ives Says Apple-Meta AI Partnership Could Help Unlock $75 Per Share in AI Value for AAPL Stock - Benzinga: 2026-10-16T11:18:12+00:00
  - Apple Inc. $AAPL Stock: Wealth Enhancement Trust Services Inc. Increases Stake - MarketBeat: 2026-10-16T08:39:31+00:00
  - Apple Inc. (AAPL): Can Longer iPhone Ownership Still Produce More Profit per Customer? - Yahoo Finance: 2026-10-15T13:38:15+00:00
  - Apple Inc. (AAPL) is Attracting Investor Attention: Here is What You Should Know - Yahoo Finance: 2026-10-13T12:00:02+00:00
- Detected Events:
  - Sold by Investors Research Corp
  - Sold by Persium Advisors LLC
  - Wealth Enhancement Trust Services Inc. Increases Stake
  - Partnership with Meta
  - Can Longer iPhone Ownership Still Produce More Profit per Customer?
  - Attracting Investor Attention
- Routed Specialist: Market
- Evaluation: Evaluation score: 82.5 Evaluation scope: Evaluation issues:
  - Evaluate the impact of market performance, financial condition, SEC evidence, and recent news on the company's current risk picture.

1.15 Risks and Uncertainties

- Risks: None provided
- Missing Information: None provided
- Follow-up Questions:
  - Include at least two evidence-based observations
  - Verify the linked sources for news articles
  - Use provider-supplied data for market performance and news
  - Consider using multiple sources to verify financial metrics and balance-sheet data

1.16 Data Quality

- Data-quality problems: None provided
- Improvements for future runs:
  - Include at least two evidence-based observations
  - Verify the linked sources for news articles
  - Use provider-supplied data for market performance and news
  - Consider using multiple sources to verify financial metrics and balance-sheet data

1.17 Verification Scope

SEC facts provide primary-source evidence for selected financial metrics. Market data and news remain provider-supplied and require independent verification.

1.18 Verified Financial Metrics

The following values are rendered deterministically from provider observations; no model arithmetic is used.

- **Current Price**: $343.46
- **Market Capitalization**: $5,012,516,765,696
- **Enterprise Value**: $5,034,461,888,512
- **Start Price**: $251.36
- **Latest Price**: $343.46
- **One-Month Return**: 1.92%
- **Three-Month Return**: 5.26%
- **One-Year Return**: 36.64%
- **Annualized Volatility**: 24.40%
- **Maximum Drawdown**: -13.80%
- **52-Week High**: $348.15
- **52-Week Low**: $242.76
- **Latest Volume**: 59,895,300
- **Average Volume**: 48,103,449
- **Price Observation Count**: 251
- **Trailing P/E**: 39.208x
- **Forward P/E**: 35.860x
- **Price/Sales**: 10.738x
- **Profit Margin**: 27.62%
- **Operating Margin**: 32.62%
- **Return on Equity**: 148.75%

- **Beta**: 1.069x
- **Revenue** (2025-09-30 00:00:00): $416,161,000,000
- **Operating Income** (2025-09-30 00:00:00): $133,050,000,000
- **Net Income** (2025-09-30 00:00:00): $112,010,000,000
- **EBITDA** (2025-09-30 00:00:00): $144,748,000,000
- **Diluted EPS** (2025-09-30 00:00:00): $7.46
- **Operating Cash Flow** (2025-09-30 00:00:00): $111,482,000,000
- **Free Cash Flow** (2025-09-30 00:00:00): $98,767,000,000
- **Capital Expenditure** (2025-09-30 00:00:00): -$12,715,000,000
- **EBITDA / Net Income** (2025-09-30 00:00:00): 1.292x
- **Dividend Yield**: 0.32% (Yahoo Finance supplies this field in percentage points.)
- **Cash and Short-Term Investments** (2025-09-30): $54,697,000,000
- **Total Debt** (2025-09-30): $98,657,000,000
- **Net Debt** (2025-09-30): $43,960,000,000
- **Total Assets** (2025-09-30): $359,241,000,000
- **Total Liabilities** (2025-09-30): $285,508,000,000
- **Stockholders' Equity** (2025-09-30): $73,733,000,000
- **Cash / Debt** (2025-09-30): 0.554x
- **Revenue Growth** (2024-09-30 to 2025-09-30): 6.43%
- **Net Income Growth** (2024-09-30 to 2025-09-30): 19.50%
- **Diluted EPS Growth** (2024-09-30 to 2025-09-30): 22.70%
- **Cash Trend** (2024-09-30 to 2025-09-30): -16.07%
- **Debt Trend** (2024-09-30 to 2025-09-30): -7.48%
- **Assets Trend** (2024-09-30 to 2025-09-30): -1.57%
- **Liabilities Trend** (2024-09-30 to 2025-09-30): -7.31%
- **Equity Trend** (2024-09-30 to 2025-09-30): 29.47%

1.19 Evidence-Based Interpretation

Because the one-year return was 36.64% while annualized volatility was 24.40% and maximum drawdown was -13.80%, the historical return must be considered together with measured price risk; these observations do not predict future performance.

Revenue growth of 6.43% and net-income growth of 19.50% indicate the latest year-over-year statement changes. Cash of $54,697,000,000 compared with total debt of $98,657,000,000 shows the reported balance-sheet exposure; however, these period snapshots alone do not establish future liquidity or earnings durability.

SEC EDGAR supplied 8 official company fact(s) and 8 recent filing record(s), providing primary-source corroboration; however, filing periods and provider statement periods should be compared explicitly before drawing conclusions.

Yahoo Finance reports stockholders' equity of $73,733,000,000 for 2025-09-30, while SEC company facts report $107,520,000,000 for fiscal 2026 Q3 ended 2026-06-27. These values describe different reporting periods and provider snapshots and should not be treated as the same balance-sheet date.

The news workflow processed 8 article(s). The structural completeness and source-traceability rubric averaged 100.00%. The separate substantive financial-reasoning and event-significance score averaged 50.00%. A 100% structural score means the expected fields and traceable source evidence were present; it does not establish perfect financial analysis. Because provider payloads may contain

only headlines or snippets, detected events remain uncertain and should be verified at the linked publishers.

Taken together, the market measurements, financial condition, SEC evidence, and current news indicate a mixed, evidence-bounded risk picture rather than a forecast or an investment recommendation.

1.19.1 8.1 Deterministic report view

This second rendering guarantees that the core company overview, price performance, valuation, financial performance, cash flow, dividend, SEC filing evidence, news-source coverage, validation findings, and missing information remain visible even if model prose varies between runs.

```
======================================================================
FINAL RESEARCH REPORT
======================================================================

Apple Inc. (AAPL)
----------------------------------------------------------------------

1. COMPANY OVERVIEW
   Sector:              Technology
   Industry:            Consumer Electronics
   Country:             United States
   Current Price:       $343.46
   Market Cap:          $5.013T
   Enterprise Value:    $5.034T

2. PRICE PERFORMANCE
   Start Price:         $251.36
   Latest Price:        $343.46
   One-Year Return:     36.64%
   Annualized Volatility: 24.40%
   Maximum Drawdown:    -13.80%
   Observations:        251

3. VALUATION
   Trailing P/E:        39.21
   Forward P/E:         35.86
   Price/Sales (TTM):   10.74
   Profit Margin:       27.62%
   Operating Margin:    32.62%
   Return on Equity:    148.75%
   Beta:                1.069

4. FINANCIAL PERFORMANCE (2025)
   Revenue:             $416.161B
   Operating Income:    $133.050B
   Net Income:          $112.010B
```

```
   EBITDA:              $144.748B
   Diluted EPS:         $7.46
   EBITDA / Net Income: 1.292x

5. CASH FLOW (2025)
   Operating Cash Flow: $111.482B
   Free Cash Flow:      $98.767B
   Capital Expenditure: -$12.715B

6. BALANCE SHEET AND GROWTH
   Cash:                $54.697B
   Total Debt:          $98.657B
   Net Debt:            $43.960B
   Cash / Debt:         0.554x
   Total Assets:        $359.241B
   Total Liabilities:   $285.508B
   Stockholders' Equity: $73.733B
   Revenue Growth:      6.43%
   Net Income Growth:   19.50%
   Diluted EPS Growth:  22.70%
   Cash Trend:          -16.07%
   Debt Trend:          -7.48%
   Assets Trend:        -1.57%

7. SEC EDGAR EVIDENCE
   Source:               SEC EDGAR
   Entity:               Apple Inc.
   CIK:                  0000320193
   SIC:                  3571 - Electronic Computers
   Official Company Facts:
      Revenue: 364,357,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27;
filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Net Income: 101,464,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27;
filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Operating Income: 122,432,000,000.00 USD (FY 2026; period Q3; ended
2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Operating Cash Flow: 116,996,000,000.00 USD (FY 2026; period Q3; ended
2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Assets: 383,266,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27;
filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Liabilities: 275,746,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27;
filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Stockholders Equity: 107,520,000,000.00 USD (FY 2026; period Q3; ended
2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Diluted EPS: 6.88 USD/shares (FY 2026; period Q3; ended 2026-06-27; filed
2026-07-31; form 10-Q; accession 0000320193-26-000020)
   Recent Filings:
      - 10-Q | filed 2026-07-31 | report 2026-06-27 | accession
```

0000320193-26-000020 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/aapl-20260627.htm
      - 8-K | filed 2026-07-30 | report 2026-07-30 | accession 0000320193-26-000018 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000018/aapl-20260730.htm
      - 10-Q | filed 2026-05-01 | report 2026-03-28 | accession 0000320193-26-000013 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000013/aapl-20260328.htm
      - 8-K | filed 2026-04-30 | report 2026-04-30 | accession 0000320193-26-000011 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000011/aapl-20260430.htm
      - 8-K | filed 2026-04-20 | report 2026-04-17 | accession 0001140361-26-015711 | https://www.sec.gov/Archives/edgar/data/320193/000114036126015711/ef20071035_8k.htm
      - 8-K | filed 2026-02-24 | report 2026-02-24 | accession 0001140361-26-006577 | https://www.sec.gov/Archives/edgar/data/320193/000114036126006577/ef20060722_8k.htm
      - 10-Q | filed 2026-01-30 | report 2025-12-27 | accession 0000320193-26-000006 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000006/aapl-20251227.htm
      - 8-K | filed 2026-01-29 | report 2026-01-29 | accession 0000320193-26-000005 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000005/aapl-20260129.htm

8. DIVIDEND
   Dividend Yield: 0.32%

9. RECENT NEWS AND CATALYSTS
   Retrieved At: 2026-10-17T05:02:47.807462+00:00
   Pipeline Stages: ingest → preprocess → classify → extract → summarize → route → specialist → evaluate → optimize
   Articles Processed: 8
   Retrieved Counts: Yahoo Finance=0, NewsAPI=0, Google News RSS=8
   Source Counts: Google News RSS=8
   Routing Counts: market=6, news=2
   Avg Structural Score: 100.0/100
   Avg Substantive Score: 50.0/100
   Passed Evaluations: 8
   Score Threshold: 80.0/100
   Optimizer Limit: 3
   Score Meaning: Structural scores measure completeness and traceability, not perfect financial analysis.
   Article Details: Saved in the companion audit HTML artifact.

10. RISKS AND UNCERTAINTIES
   - Substantive financial reasoning or event-significance analysis remains limited

```
11. DATA QUALITY
   - Evidence strength: Evidence coverage
   - Evidence strength: Data-quality problems
   - Evidence strength: Improvements for future runs

12. MISSING INFORMATION
   None

13. FURTHER RESEARCH
   - Include at least two evidence-based observations
   - Verify the linked sources for news articles
   - Use provider-supplied data for market performance and news
   - Consider using multiple sources to verify financial metrics and balance-
sheet data
   - What is the current market performance of AAPL and how does it impact the
company's risk picture?
   - How does the company's financial condition (e.g., cash flow, debt) reflect
its current risk profile?
   - What SEC evidence (e.g., EDGAR filings) can be used to inform the company's
risk assessment?
   - What recent news stories or events may be relevant to the company's current
risk picture?
   - Which material claims can be linked to filing-level citations or provider
source URLs?

======================================================================
END OF FINAL RESEARCH REPORT
======================================================================
```

1.20 9. Ten-year market context

The original notebook's 10-year graph is preserved as a reusable reporting function. It uses adjusted closing prices and reports the start price, end price, and total return.



```
10-year price summary: $26.88 → $343.46 (+1,177.7%)
```

1.21 10. Evaluation evidence and iteration

Each news analysis receives a **structural completeness and source-traceability score** with 25 points for each of the following:

- at least one concrete observation;
- a clear interpretation;
- an explicit uncertainty or missing-information statement; and
- at least two evidence-based observations.

A separate substantive score checks whether the observations go beyond headline/source metadata and whether the interpretation explains financial reasoning or event significance. Thus, 100/100 on the structural rubric establishes expected fields and traceability—not perfect financial analysis. Below-threshold structural work is revised up to `MAX_OPTIMIZER_ITERATIONS`. The complete report is separately scored for completeness, grounding, reasoning, consistency, sources, and clarity; LangGraph routes a failed draft through the report optimizer and back to evaluation up to `MAX_REPORT_REVISIONS`. Before/after scores and every graph node are retained below.

1.21.1 Complete-report evaluator–optimizer history

	evaluation	score	passed	completeness	grounding	reasoning	consistency	sources	clarity
0	0	82.5	False	100.0	100.0	15.0	100.0	80.0	100.0
1	1	90.0	True	100.0	100.0	60.0	100.0	80.0	100.0


```
{
  "report_revisions": [
    {
      "iteration": 1,
      "before_score": 82.5,
      "after_score": 90.0,
      "feedback": [
        "Improve reasoning; current score is 15/100.",
        "No report can pass while these criteria are below 60/100: reasoning."
      ]
    }
  ],
  "graph_trace": [
    "plan_request",
    "comprehensive_research",
    "validate_observations",
    "reflect",
```

```
    "synthesize",
    "evaluate_report",
    "revise_report",
    "evaluate_report",
    "finalize"
  ]
}
```

1.21.2 Per-article evaluator–optimizer history

	article	revision	structural_score	structural_passed	\
0	1	0	100	True	
1	2	0	0	False	
2	2	1	100	True	
3	3	0	100	True	
4	4	0	100	True	
5	5	0	100	True	
6	6	0	0	False	
7	6	1	100	True	
8	7	0	100	True	
9	8	0	100	True	

	substantive_reasoning_score	\
0	50	
1	0	
2	50	
3	50	
4	50	
5	50	
6	0	
7	50	
8	50	
9	50	

	feedback
0	Structural completeness passed, but substantiv…
1	Add concrete observations from the article.; A…
2	Structural completeness passed, but substantiv…
3	Structural completeness passed, but substantiv…
4	Structural completeness passed, but substantiv…
5	Structural completeness passed, but substantiv…
6	Add concrete observations from the article.; A…
7	Structural completeness passed, but substantiv…
8	Structural completeness passed, but substantiv…
9	Structural completeness passed, but substantiv…

1.21.3 Requirement-specific behavioral demonstration

```
                   requirement  \
0          market-only routing
1       financial-only routing
2            news-only routing
3        comprehensive routing
4    weak complete-report test
5  strong complete-report test
6          two-run memory test

                                          expected  \
0                                           market
1                                        financial
2                                             news
3                                     comprehensive
4                                             fail
5                                             pass
6  Run 2 retrieves feedback and adds a filing/sou…

                                            actual  passed
0                                           market    True
1                                        financial    True
2                                             news    True
3                                     comprehensive    True
4                                             fail    True
5                                             pass    True
6  memory_retrieved=True; traceability_question_a…    True
```

1.21.4 Behavioral and regression test results

```
… [ 66%]
…s…               [100%]
100 passed, 1 skipped, 30 subtests passed in 2.71s
pytest exit code: 0
```

1.22 11. Memory and learning across runs

The first run stores concise lessons and evaluator feedback in an isolated demonstration memory file; the second run receives up to the last three ticker-relevant entries in its planner prompt. The comparison below separates successful retrieval from substantive planning improvement. Source-traceability feedback causes the planner to add a filing/source-citation question deterministically; if that trigger is absent, the notebook reports retrieval without claiming learning.

```
                                  evidence  \
0  Saved feedback records received by Run 2
1                           Selected tools
2                  Added planning questions
3       Filing/source traceability question
```

```
                                                run_1  \
0                                                   0
1  balance_sheet, cash_flow, company_info, financ…
2                                                   -
3                                               False

                                                run_2  changed
0                                                   1     True
1  balance_sheet, cash_flow, company_info, financ…    False
2  What is the current market performance of AAPL…     True
3                                                True     True
```

```
Saved Run 1 feedback:
{
  "date": "2026-10-17T05:02:14.249666+00:00",
  "symbol": "AAPL",
  "summary": "Here are the reusable research lessons from this AAPL run:\n\n1.
**Evidence coverage**: Ensure that primary-source evidence (e.g., SEC EDGAR) is
used to verify financial metrics, but be cautious when comparing data from
different providers and reporting periods.\n2. **Data-quality problems**: Verify
data quality by cross-checking values from different providers and reporting
periods to avoid misinterpretation.\n3. **Improvements for future runs**:\n*
Include at least two evidence-based observations to strengthen the analysis.\n*
Verify the linked sources for news articles to ensure accuracy.\n* Use provider-
supplied data for market performance and news, but be cautious of potential
biases and limitations.\n* Consider using multiple sources to verify financial
metrics and balance-sheet data.\n* Improve financial reasoning and event-
significance analysis to strengthen the analysis.\n* Consider using more
advanced analytics and modeling techniques to better understand the company's
risk picture.",
  "weaknesses": [
    "Substantive financial reasoning or event-significance analysis remains
limited"
  ]
}
Learning interpretation: Run 1 feedback identified weak source traceability. Run
2 retrieved that feedback and added a question requiring filing-level citations
or provider source links. The selected tools did not change because the
comprehensive route already included the necessary SEC and provider tools.
```

1.23 12. Limitations, responsible use, and conclusion

- Yahoo Finance, NewsAPI, Google News RSS, and SEC EDGAR can be delayed, unavailable, rate-limited, incomplete, or inconsistent.
- News summaries may contain only provider snippets; source links should be reviewed before decisions.
- Keyword extraction and routing are intentionally transparent but cannot capture every nu-

ance.

- Local-model output is nondeterministic and may still omit context; deterministic validation reduces but cannot eliminate that risk.
- Google News RSS is a coverage fallback, not a replacement for primary or official evidence.
- FRED and Alpha Vantage remain scaffolds and are not represented as report evidence.
- Historical performance, evaluator scores, and model interpretations do not predict future returns.

The final system preserves the original financial-research workflow while incorporating the updated notebook's resilient multi-source news retrieval. Its modular adapters, explicit state, provenance, bounded revision loop, deterministic report, persistent memory, offline tests, and generated diagrams make the work reproducible and auditable.

1.24 Appendix — companion audit artifact

The complete workflow state is written to a companion HTML file rather than repeated in the main report. This preserves every plan, observation, article, evaluator result, revision, graph node, and memory artifact while keeping the review copy compact.

	artifact	record_count
0	symbol	1
1	research_request	1
2	remember	1
3	request_route	1
4	selected_specialists	3
5	plan	7
6	observations	8
7	specialist_analyses	3
8	news_analysis	16
9	validation	0
10	reflection	6
11	report	1
12	draft_report	1
13	best_report	1
14	best_report_evaluation	7
15	report_validation	0
16	report_evaluation	7
17	report_evaluation_history	2
18	report_revision_history	1
19	report_revision_count	1
20	max_report_revisions	1
21	graph_trace	9

```
Complete audit artifact: /Users/brandonwirgau/Projects/AAI/520/Final
Project/notebooks/multi_agent_financial_analysis_audit.html
Audit size: 176,360 bytes
```